The Bellman Equation: A Comprehensive Guide

A Step-by-Step Derivation with Practical Examples

Abstract

This document provides a complete derivation of the Bellman Equation using a **single running example**: a pizza delivery driver navigating a city. Every concept is illustrated with concrete numbers you can trace through by hand. By the end, you'll understand not just *what* the Bellman Equation says, but *why* it must be true.

I. Part 1: The Big Picture

A. The Problem We're Solving

You're a pizza delivery driver. You have a map of the city with different intersections. Your goal: figure out which route maximizes your tips over your entire shift.

The Core Question

If I'm standing at this intersection right now, how much money will I make from here until I clock out?
This number—the total expected future earnings from a location—is called the **Value** of that location.

Here's the insight that makes the Bellman Equation powerful:

You don't need to simulate your entire future to know how good your current spot is. You only need to know:

1) What you'll earn in the **next few minutes** (immediate reward)
2) How good the **next intersection** is (and you can look that up!)

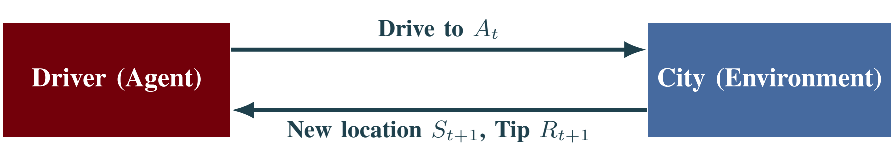


Fig. 1. The RL Loop: You (the agent) pick a route. The city (environment) gives you a new location and a tip.

II. Part 2: The Vocabulary with Numbers

We'll define every term using our pizza delivery scenario, with **actual numbers** you can follow.

A. State: s — "Where Am I?"

Pizza Delivery Map

Imagine the city has 4 key locations:

- s_A = Downtown (lots of apartments, good tips)
- s_B = Suburbs (far drives, okay tips)
- s_C = University (students tip poorly but order often)
- s_D = Restaurant (home base, no deliveries here)

Your state is your current location. If you're downtown, $S_t = s_A$.

B. Reward: R_{t+1} — "What Did I Just Earn?"

The reward is the **immediate payoff** after you take an action.

Concrete Tip Values

- Deliver downtown (s_A): Tip = \$8
- Deliver to suburbs (s_B): Tip = \$5
- Deliver to university (s_C): Tip = \$3
- Sitting at restaurant (s_D): Tip = \$0

Note: We write R_{t+1} (not R_t) because you receive the tip *after* completing the delivery.

C. Return: G_t — "My Total Earnings"

The **Return** is the sum of ALL future tips from now until you clock out.

A Sample Shift

Suppose your remaining shift goes: Downtown $\rightarrow$ Suburbs $\rightarrow$ University $\rightarrow$ Done.
Your tips would be: \$8 + \$5 + \$3 = \$16
So if you're currently at the restaurant about to start this route:

$$G_t = R_{t+1} + R_{t+2} + R_{t+3} = 8 + 5 + 3 = 16$$

D. Discount Factor: γ — "A Dollar Today vs. Tomorrow"

Here's a twist: money **now** is worth more than money **later**. Why?

- You might get in an accident and never collect later tips
- You could invest today's money
- There's uncertainty about the future

We capture this with γ (gamma), a number between 0 and 1.

The \$100 Test

Would you rather have:

(A) \$100 right now, or

(B) \$100 one hour from now?

Most people pick (A). If you're *indifferent* between \$100 now and \$X in an hour, then:

$$\gamma = \frac{100}{X}$$

If you need \$111 later to feel indifferent, then $\gamma = 100/111 \approx 0.9$.

Discounted Return Formula

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 R_{t+4} + \cdots$$

With $\gamma = 0.9$ and tips of \$8, \$5, \$3:

$$\begin{aligned} G_t &= 8 + (0.9)(5) + (0.9)^2(3) \\ &= 8 + 4.50 + 2.43 \\ &= \$14.93 \end{aligned}$$

Notice: The \$3 tip (two steps away) only contributes \$2.43 to today's value.

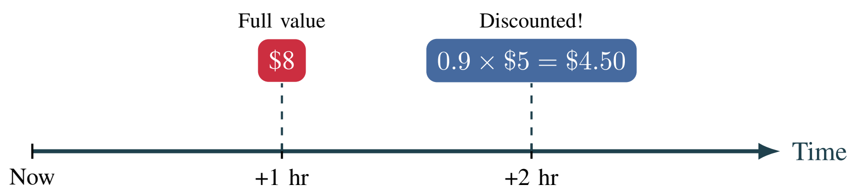


Fig. 2. Future rewards shrink. A \$5 tip one hour from now is only worth \$4.50 in "today dollars."

E. *Value Function: $v(s)$ — "How Good Is This Spot?"*

Definition of Value

$$v(s) = \mathbb{E}[G_t \mid S_t = s]$$

Translation: "The average total money I'll make if I start at location s."

Intuition Check

Which location has higher value?

- Downtown (s_A): High tips, more orders nearby
- University (s_C): Low tips, but you might get lucky

Clearly $v(s_A) > v(s_C)$. The Bellman equation will let us compute *exactly* how much higher.

III. Part 3: The Math Tools You Need

A. Expectation $\mathbb{E}[\cdot]$ = Weighted Average

When outcomes are random, we compute the **average** weighted by probability.

Traffic Light Example

You're approaching an intersection.

- 70% chance: Green light $\rightarrow$ you save 2 minutes
- 30% chance: Red light $\rightarrow$ you lose 1 minute

Expected time saved:

$$\mathbb{E}[\text{time}] = (0.70)(+2) + (0.30)(-1) = 1.4 - 0.3 = +1.1 \text{ minutes}$$

You can't actually save exactly 1.1 minutes, but *on average*, this intersection helps you.

B. Linearity of Expectation

This rule is incredibly useful: **the average of a sum equals the sum of the averages**.

$$\mathbb{E}[A + B] = \mathbb{E}[A] + \mathbb{E}[B]$$

Why This Matters

If your shift has two deliveries:

- Expected tip from delivery 1: \$6
- Expected tip from delivery 2: \$4

Then your expected total is $\mathbb{E}[\text{total}] = 6 + 4 = \$10$.
You don't need to consider all combinations of outcomes!

IV. Part 4: The Derivation (With Numbers!)

We will derive the Bellman Equation in 6 steps. Here's the roadmap:

Derivation Roadmap

1) **Start** with the definition of value (total future earnings)
2) **Split** into "now" and "later"
3) **Separate** the immediate reward from the future
4) **Handle uncertainty** about where we go next
5) **Recognize** that the future value IS the value function
6) **Combine** everything into one beautiful equation

A. Setup: Our Running Example

Throughout this derivation, we'll use a concrete example with real numbers.

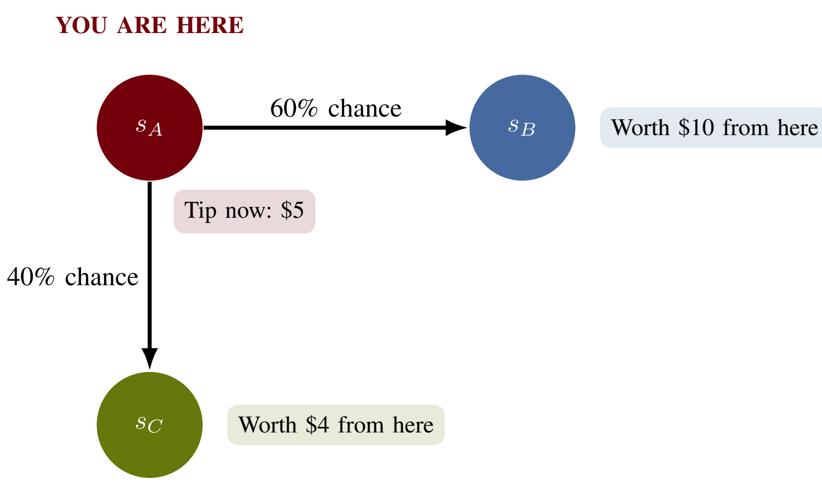


Fig. 3. Our example: You're at s_A. You get \$5 immediately, then randomly go to s_B or s_C.

The Numbers We'll Use

What we know:

Immediate tip at s_A:	$r(s_A) = \$5$
Chance of going to s_B:	$p(s_B\|s_A) = 0.6$ (60%)
Chance of going to s_C:	$p(s_C\|s_A) = 0.4$ (40%)
Value of s_B (already computed):	$v(s_B) = \$10$
Value of s_C (already computed):	$v(s_C) = \$4$
Discount factor:	$\gamma = 0.9$
What we want to find:	$v(s_A) = ???$

B. Step 1: Start with the Definition

The question we're answering: "How much money will I make in total if I start at state s?"

The value function $v(s)$ is defined as the *expected* total future earnings:

$$\boxed{v(s) = \mathbb{E}[\text{Total future earnings} \mid \text{Starting at } s] = \mathbb{E}[G_t \mid S_t = s]}$$

Remember, the total earnings G_t is the sum of all future rewards (with discounting):

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots$$

So we can write:

$$v(s) = \mathbb{E}[\underbrace{R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots}_{G_t} \mid S_t = s]$$

What This Means for s_A

We want to find: "If I'm at location s_A, what's my expected total earnings?"
This includes:

- The \$5 tip I get right now
- Plus all the (discounted) tips I'll get in the future

C. Step 2: Split "Now" vs "Later"

Here's the key trick. Let's separate the **first reward** from **everything else**:

$$v(s) = \mathbb{E}[\underbrace{R_{t+1}}_{\text{first tip}} + \underbrace{\gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots}_{\text{all future tips}} \mid S_t = s]$$

Now, factor out γ from the "future tips" part:

$$v(s) = \mathbb{E}[R_{t+1} + \gamma \underbrace{(R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \cdots)}_{\text{This looks familiar...}} \mid S_t = s]$$

The Magic Recognition

Look at $(R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \cdots)$.
This is the total return starting from time $t+1$. That's just G_{t+1}!
Pattern: $G_t = R_{t+1} + \gamma G_{t+1}$
"Today's total = today's reward + discounted tomorrow's total"

Substituting G_{t+1}:

$$v(s) = \mathbb{E}[\, R_{t+1} + \gamma G_{t+1} \mid S_t = s\,] \tag{1}$$

Pizza Delivery Intuition

Your total career earnings = Today's tip + (0.9 $\times$ your total earnings starting tomorrow)
For s_A: $v(s_A) = \$5 + 0.9 \times$ (value of wherever I end up next)

D. Step 3: Separate Immediate Reward from Future

Using the linearity property ($\mathbb{E}[A+B] = \mathbb{E}[A] + \mathbb{E}[B]$), we split the expectation:

$$v(s) = \underbrace{\mathbb{E}[R_{t+1} \mid S_t = s]}_{\text{Part A: Immediate}} + \gamma \underbrace{\mathbb{E}[G_{t+1} \mid S_t = s]}_{\text{Part B: Future}} \tag{2}$$

Part A is simple: it's just $r(s)$, the average immediate reward at state s.

Part B is trickier: what's the expected future value when we don't know where we'll go next?

Progress Check for s_A

So far we have:

$$v(s_A) = \underbrace{5}_{\substack{\text{Part A:} \\ \text{tip now}}} + 0.9 \times \underbrace{\mathbb{E}[G_{t+1} \mid S_t = s_A]}_{\substack{\text{Part B:} \\ \text{??? (need to figure this out)}}}$$

Part A: Done! It's \$5.
Part B: We need to calculate the expected future value. But we might go to s_B OR s_C...

E. Step 4: Handle the Uncertainty

Here's the problem: we don't know which state we'll land in next. It's random!

- 60% chance we go to s_B (which is worth \$10)
- 40% chance we go to s_C (which is worth \$4)

Solution: Take the **weighted average** of all possibilities.

The Averaging Principle

When you don't know which outcome will happen, **average over all possibilities**, weighted by their probabilities.

$$\text{Expected future} = \sum_{\text{all possible next states } s'} \underbrace{p(s'|s)}_{\text{probability}} \times \underbrace{(\text{value of } s')}_{\text{outcome}}$$

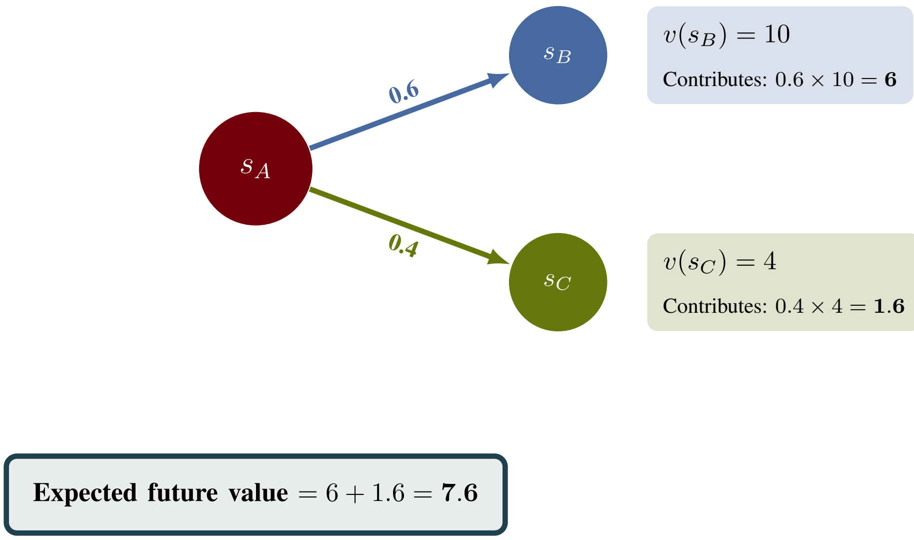


Fig. 4. The "Backup" Diagram: Average over all possible next states.

Mathematically, this is the **Law of Total Expectation**:

$$\mathbb{E}[G_{t+1} \mid S_t = s] = \sum_{s'} p(s'|s) \cdot \underbrace{\mathbb{E}[G_{t+1} \mid S_{t+1} = s']}_{\text{value if we land at } s'} \tag{3}$$

F. Step 5: Recognize the Recursion

Look closely at the term $\mathbb{E}[G_{t+1} \mid S_{t+1} = s']$ from Step 4.

Question: What does this represent?

Answer: "The expected total return starting from time $t+1$, given we're in state s' at time $t+1$."

The Aha! Moment

But wait... "expected total return starting from state s'" is **exactly the definition of** $v(s')$!

$$\mathbb{E}[G_{t+1} \mid S_{t+1} = s'] = v(s')$$

This is the recursion! The value of a state depends on the values of neighboring states.

Substituting this back into our equation from Step 4:

$$\mathbb{E}[G_{t+1} \mid S_t = s] = \sum_{s'} p(s'|s) \cdot v(s') \tag{4}$$

Computing Part B for s_A

Now we can calculate the "Part B" we left hanging in Step 3:

$$\begin{aligned}\mathbb{E}[G_{t+1} \mid S_t = s_A] &= p(s_B|s_A) \cdot v(s_B) + p(s_C|s_A) \cdot v(s_C) \\ &= (0.6)(10) + (0.4)(4) \\ &= 6 + 1.6 \\ &= \mathbf{7.6}\end{aligned}$$

The expected value of "wherever I end up next" is \$7.6.

G. Step 6: Put It All Together

Now we combine everything. From Step 3, we had:

$$v(s) = r(s) + \gamma \cdot \mathbb{E}[G_{t+1} \mid S_t = s]$$

From Step 5, we found:

$$\mathbb{E}[G_{t+1} \mid S_t = s] = \sum_{s'} p(s'|s) \cdot v(s')$$

Substituting:

The Bellman Equation

$$\boxed{v(s) = r(s) + \gamma \sum_{s'} p(s'|s) \cdot v(s')}$$

The Equation in Plain English

$$\begin{aligned}\textbf{Value of being here} &= \textbf{What I get now}\\ &+ \textbf{Discounted average of where I might go}\end{aligned}$$

Or even simpler: *"Today's value = Today's reward + Tomorrow's (discounted) value"*

Final Answer for s_A

$$\begin{aligned}v(s_A) &= r(s_A) + \gamma \cdot \big[p(s_B|s_A) \cdot v(s_B) + p(s_C|s_A) \cdot v(s_C)\big]\\ &= \underbrace{5}_{\text{tip now}} + 0.9 \times \underbrace{7.6}_{\text{expected future}}\\ &= 5 + 6.84\\ &= \boxed{\$11.84}\end{aligned}$$

Interpretation: Standing at location s_A, you can expect to earn \$11.84 total (in today's dollars).

V. Part 5: Why This Matters

A. The Power of Recursion

The Bellman equation is **recursive**: it defines $v(s)$ in terms of $v(s')$ for neighboring states.

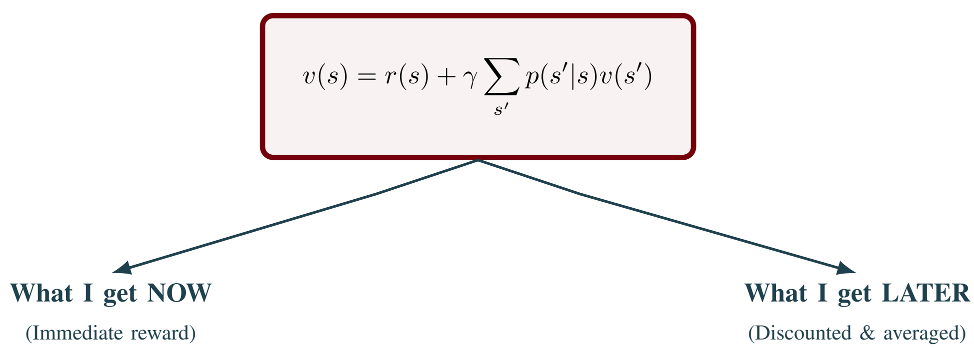
The Bootstrap Principle

You don't need to simulate the entire future.
If someone tells you the value of all neighboring states, you can immediately compute your own value using one simple formula.

B. Three Analogies to Remember

1) **GPS Navigation**: Your estimated arrival time = time to next intersection + ETA from that intersection. The GPS doesn't re-calculate the entire route; it just looks one step ahead.
2) **House Prices**: Your home's value = what you could rent it for this month + discounted future rental income. Real estate appraisers use exactly this logic.
3) **Career Planning**: Value of your current job = this year's salary + discounted value of where this job leads (promotions, opportunities).

C. The Complete Picture



Memory Hook: "Today's paycheck + tomorrow's career prospects"

Fig. 5. The Bellman Equation decomposes value into immediate and future components.

VI. Part 6: Quick Reference

Symbol	**Name**	**Pizza Delivery Analogy**
s	State	Your current location
S_t	State at time t	Where you are at hour t of your shift
R_{t+1}	Reward	The tip you receive after a delivery
$r(s)$	Expected immediate reward	Average tip when delivering from location s
G_t	Return	Total tips from now until end of shift
γ	Discount factor	How much you value future tips vs. now
$v(s)$	Value function	Expected total earnings starting from s
$p(s'\|s)$	Transition probability	Chance of going to s' when leaving s

VII. Conclusion

The Bellman Equation tells us something profound:

The One Sentence Summary

The value of where you are = what you get now + discounted average of where you might go next.

This recursive structure is what makes reinforcement learning tractable. Instead of computing infinite futures, we break the problem into manageable one-step lookaheads.

Next time you use a GPS, think about how it estimates your arrival time: it doesn't simulate every possible traffic scenario for your entire route. It just looks at the next segment and trusts that the rest has already been computed. That's the Bellman equation in action.